INCH-POUND

MIL-C-24640/24C
26 June 1995
SUPERSEDING
MIL-C-24640/24B

MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT, MULTI-PAIR, AWG 22, TYPE TTXW (INCLUDING VARIATION TYPES TTXOW AND TTXWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

CONSTRUCTION
(Watertight)

FIRST - Copper conductor, uncoated, AWG 22, 7 strands, in accordance with ASTM B-8. Nominal strand diameter 0.031 inch.

SECOND - Mica glass tape insulation 0.0029 ±.0002 inch (50 percent overlap maximum and minimum overlap of 45 percent).

THIRD - Opaque-pigmented polyimide tape .001 inch (50 percent overlap minimum).

FOURTH - Opaque-pigmented polyimide tape .001 inch with 0.0001 inch Fluorinated Ethylene Propylene [FEP] (50 percent overlap minimum).

FIFTH - Modified aromatic polyimide resin coating 0.0005 inch minimum (.C01 inch nominal). Identification codes applied using Method 3 (telephone color code) or 6. (Note for method 6, Twisted pair identification code colors shall be one black and one white conductor. For type TTXW (A)(OW) 1 1/2, colors shall be one white one black and one red conductor.

SIXTH - Two conductors twisted together. (Except for type TTXW (A)(OW) 1 1/2, shall be 3 conductors twisted together to form a triad.) The pitch shall be 1½" ± ¼" (inch).

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SEVENTH - The required number of component wires shall be cabled together. Fillers may be used as necessary, to form a round cable.

EIGHTH - Binder tape applied helically with overlap.

CONSTRUCTION VARIANT FOR TYPE TTXW

NINTH - Crosslinked polyolefin jacket. (See Table I for thickness).

CONSTRUCTION VARIANT FOR TYPE TTXWA

NINTH - Crosslinked polyolefin jacket. (See Table II for thickness).

TENTH - Braided metal armor. Dimensions in accordance with Table II.

CONSTRUCTION VARIANT FOR TYPE TTXOW

NINTH - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface transfer Impedance requirements as listed herein.

TENTH Binder tape applied helically with overlap.

ELEVENTH - Crosslinked polyolefin jacket. (See Table III for thickness).

TABLE I. Construction details - unarmored.

PIN M24640/24	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UN	TTXW-1-1/2	1-1/2	0.028	0.181	0.195	26.0
-02UN	TTXW-3	3	.038	.285	.307	62.3
-03UN	TTXW-5	5	.037	.331	.357	91.0
-04UN	TTXW-10	10	.041	.456	.492	172.0
-05UN	TTXW-15	15	.044	.527	.569	235.0
-06UN	TTXW-20	20	.044	.577	.621	285.0
-07UN	TTXW-30	30	.044	.684	.738	417.0
-08UN	TTXW-40	40	.047	.790	.852	544.0

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TABLE II. Construction details - armored.

PIN M24640/24	Type number	Number of conductors	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01AN	TTXWA-1-1/2	1-1/2	0.028	0.231	0.245	41.0
-02AN	TTXWA-3	3	.038	.335	.357	84.3
-03AN	TTXWA-5	5	.038	.381	.407	119.4
-04AN	TTXWA-10	10	.041	.506	.542	209.3
-05AN	TTXWA-15	15	.044	.577	.619	269.6
-06AN	TTXWA-20	20	.044	.627	.671	328.2
-07AN	TTXWA-30	30	.044	.734	.788	463.2
-08AN	TTXWA-40	40	.047	.840	.902	592.6

TABLE III. Construction details - overall shield.

PIN M24640/24	Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UO	TTXOW-1-1/2	1-1/2	0.127	0.031	0.253	0.273	67.9
-02UO	TTXOW-3	3	.216	.038	.333	.359	108.0
-03UO	TTXOW-5	5	.263	.038	.376	.406	134.0
-04UO	TTXOW-15	15	.449	.044	.556	.600	299.0
-05UO	TTXOW-20	20	.510	.044	.614	.662	371.0
-06UO	TTXOW-30	30	.610	.047	.717	.772	497.0
-07UO	TTXOW-40	40	.720	.047	.823	.887	631.0

REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

Finished wire:

For polyimide tape constructions, the following additional tests shall be applicable in accordance with MIL-W-81381:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Blocking: Oven temperature, 200± 2°C (392± 3.6°F) no sticking of conductors
Flammability: 3 seconds (max) after flame.
3.0 inches (max) flame travel.
No flaming of tissue paper.

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Humidity resistance: 5 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test: 100 percent test. Impulse voltage at 8.0 kV.
Insulation resistance: 2500 megohms for 1000 feet (min).
Polyimide cure test: No cracking.
Shrinkage: 0.062 inch maximum at 180°C
Surface resistance: 5 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F). Maximum change in measurement: 0.031 inch.
Wet dielectric test: 2500 volts (rms).

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.54
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield (TTXOW only)	1000
Shield to shield	Not applicable
* Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor	2500
Conductor to shield (TTXOW only)	2500
Shield to shield (component shield only)	Not applicable
Shield to shield (component shield to overall shield	Not applicable
Conductor and shield continuity	No failure
Jacket flaws (type TTXOW only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
Capacitance unbalance - percent, maximum	Not applicable
Characteristic impedance at 1 MHz, ohms	Not applicable
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
Weight	No failure
Group B:	
Cold bending	No failure
Drip	No failure
* Cross link proof test (percent maximum)	
Jacket	50
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160

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	Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
	Gas flame (1 hour)	No failure
	Group C:	
	Physicals (aged)	
	Jacket	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged, minimum ...	60
	Hot oil immersion	
*	Tensile strength - percent of unaged, ... minimum	50
*	Elongation - percent of unaged minimum	50
*	Permanence of printing (conductor) - Method 6 only cycles, minimum	25
*	Permanence of printing (jacket) - cycles, minimum ...	125
	Armor - conformance to material, (type TTXWA only) construction and coverage	No failure
	Shield - conformance to material, construction and coverage (type TTXOW only)	No failure
	Heat distortion - percent of unaged, maximum.....	30
*	Cable sealant removability.....	No failure
	Shrinkage.....	No failure
	Mutual inductance - dB/1000 feet, maximum	Not applicable
*	Surface transfer impedance (type TTXOW only)	
	Milliohms per meter, maximum	700
	EMP response, dB, minimum	60
	Pulse response time - ms/500 feet	
	Rise time, maximum	Not applicable
	Decay time, maximum	Not applicable
*		
	Group D	
	Flame propagation	No Failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket.....	2
Fillers.....	2
Insulation.....	18
Halogen content - percent, maximum	
Jacket.....	0.2
Fillers.....	0.2

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	Immersion	
	Tensile strength - percent of unaged, minimum...	50
	Elongation - percent of unaged, minimum	50
*	Smoke index, maximum	
	Jacket.....	25
	Fillers.....	45
	Insulation.....	10
	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure
	Fungus Resistance	No failure
	Abrasion resistance	No failure
	Gas Flame (3 hour)	No failure
	Component wire	No failure

Unit ordering length: 1 1/2 - 15 500 feet nominal.
20 - 40 250 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - SH

Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
DLA - IS

Preparing Activity:
Navy - SH

Agent:
DLA-IS

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